Neural NLP Pipeline for Urdu Text

Detailed Assignment 2 Report

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Course: CS-4063 Natural Language Processing
Framework: PyTorch from scratch

GITHUB: 22i0452/i22-0452-NLP-Assignment2

Corpus	Vocab	Outputs
50 cleaned docs	671 tokens	DOCX + PDF
50 metadata labels	Word2Vec available	Figures embedded

Executive summary The full pipeline was implemented successfully. Part 1 produced TF-IDF, PPMI and Skip-gram outputs; Part 2 produced meaningful POS results but very weak NER due to sparse labels; Part 3 produced the strongest model-level result, with Transformer test accuracy 0.70 and macro-F1 0.6267 on 5-way topic classification.

Overview

Objective: Build a complete neural NLP pipeline for Urdu text using only from-scratch PyTorch components. The pipeline covered weighted lexical representations, predictive embeddings, sequence labeling, and document classification.

Key run summary

Part	Main output	Best metric	Status
Part 1	TF-IDF, PPMI, Word2Vec	W2V loss 7.50 -> 3.42	Done
Part 2	BiLSTM POS + NER	POS test acc 0.7103	Done
Part 3	Transformer classifier	Test acc 0.7000	Done

Part 1 Results: TF-IDF, PPMI, and Skip-gram Word2Vec

- Data used: 50 cleaned documents and a vocabulary of 671 tokens.

- TF-IDF matrix saved successfully with shape (50, 671). The extracted topic-specific words were sensible overall: politics emphasized اعلان, اپوزیشن, بحث, اجلاس; sports emphasized کھلاڑیوں, کوچ, میچ, ٹیم; economy emphasized سرمایہ, لاگت, قیمتوں; health & society emphasized پانی, بیماریوں, صحت.

- PPMI matrix was built with window size k = 5 and saved with shape (671, 671). The t-SNE projection was generated for top frequent words. Rendering of Urdu labels was affected by font limitations, but the plot still demonstrates spread of the learned PPMI space.

• Skip-gram Word2Vec trained correctly for 5 epochs. Training loss decreased as 7.4956 -> 6.2756 -> 4.4865 -> 3.6495 -> 3.4195, which confirms that optimization was working and the model learned non-trivial structure.

• Embedding quality remained limited because the corpus was small and several required evaluation words were missing or weakly represented. In the 4-condition comparison, PPMI was the most stable baseline under low-data conditions.

Part 1 evaluation snapshot

Measure	Observed output
PPMI neighbours for پاکستان	کھیلے, کرکٹ, گئے, لاہور, بورڈ
PPMI neighbours for صحت	بنیادی, غذائیت, محکمہ, گھر, سہولیات
MRR C1 / C2 / C3 / C4	0.0947 / 0.0458 / 0.0050 / 0.0832
Saved embedding matrix	(671, 100)

Figure 1. PPMI t-SNE of frequent words

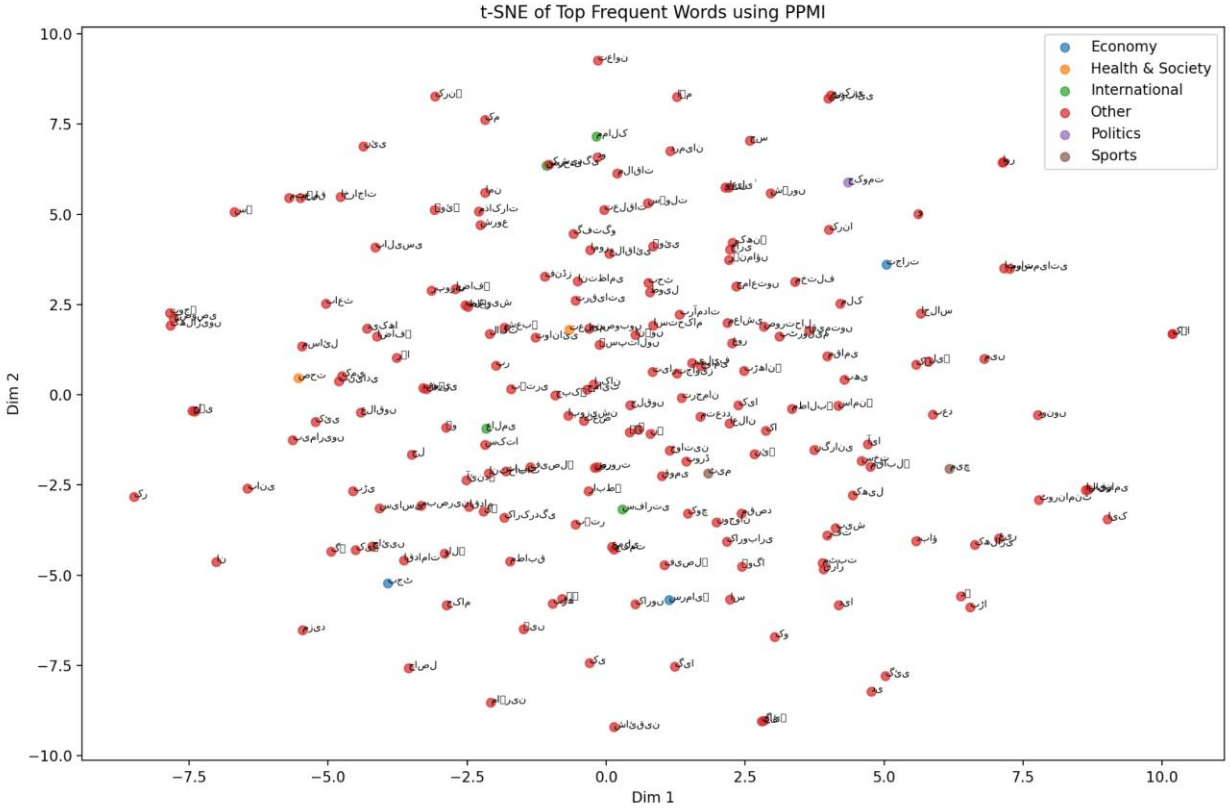
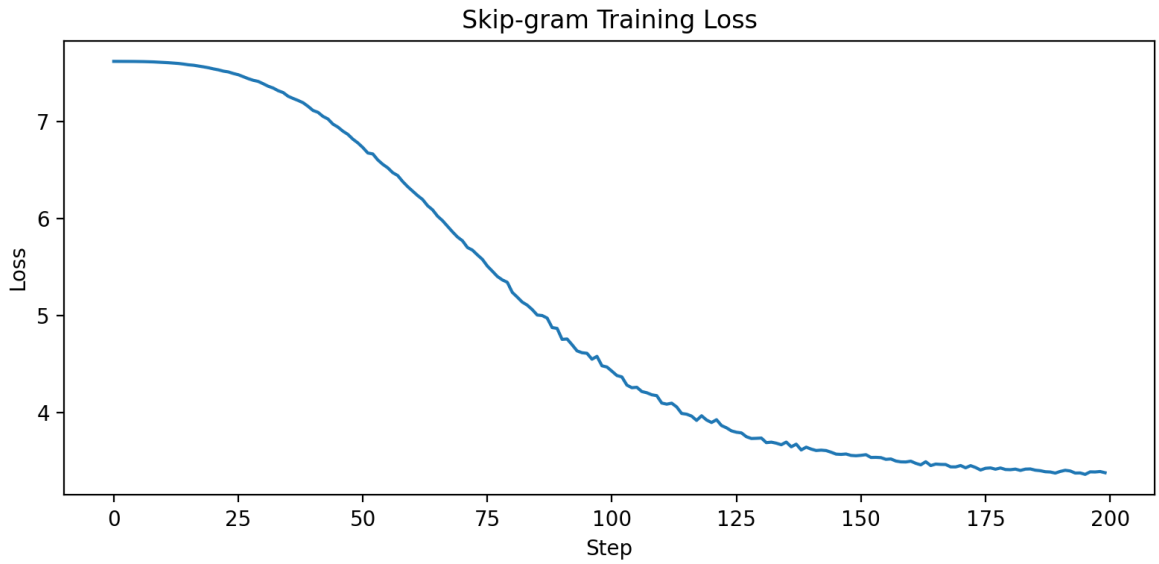


Figure 2. Skip-gram Word2Vec training loss



Part 2 Results: BiLSTM POS Tagging and NER

• Sentence extraction produced 150 usable sentences from the available corpus. This is below the assignment target of 500, so the run should be interpreted as a reduced but fully functional demonstration pipeline.

• The split was balanced by topic: train 105, validation 20, and test 25. POS labels were dominated by NOUN, VERB, POST, and ADJ, while NER labels were overwhelmingly O, which strongly affected downstream NER learning.

• POS tagging worked reasonably well. The frozen model reached test accuracy 0.4891 and macro-F1 0.1333, while the fine-tuned model reached test accuracy 0.7103 and macro-F1 0.4679. This is the clearest evidence that fine-tuning the Part 1 embeddings helped.

• The biggest POS confusions were NOUN -> ADJ, NOUN -> VERB, and ADJ -> NOUN. This is plausible in Urdu news text because many words can behave descriptively or nominally depending on context and because the POS supervision was rule-based rather than manually gold-labeled.

• NER failed to generalize. Both BiLSTM-CRF and BiLSTM-softmax produced zero precision, recall, and F1 on the test set because positive entity labels were too sparse. Most sequence positions were labeled O, so the model had almost no signal for learning robust entity boundaries.

Part 2 metrics and ablations

Experiment	Metric	Value
POS frozen	Test macro-F1	0.1333
POS fine-tuned	Test macro-F1	0.4679
NER + CRF	Entity F1	0.0000
NER without CRF	Entity F1	0.0000
A1 Unidirectional	POS val F1	0.0612
A2 No dropout	POS val F1	0.1320
A3 Random embeddings	POS val F1	0.1387

Figure 3. POS train/validation loss

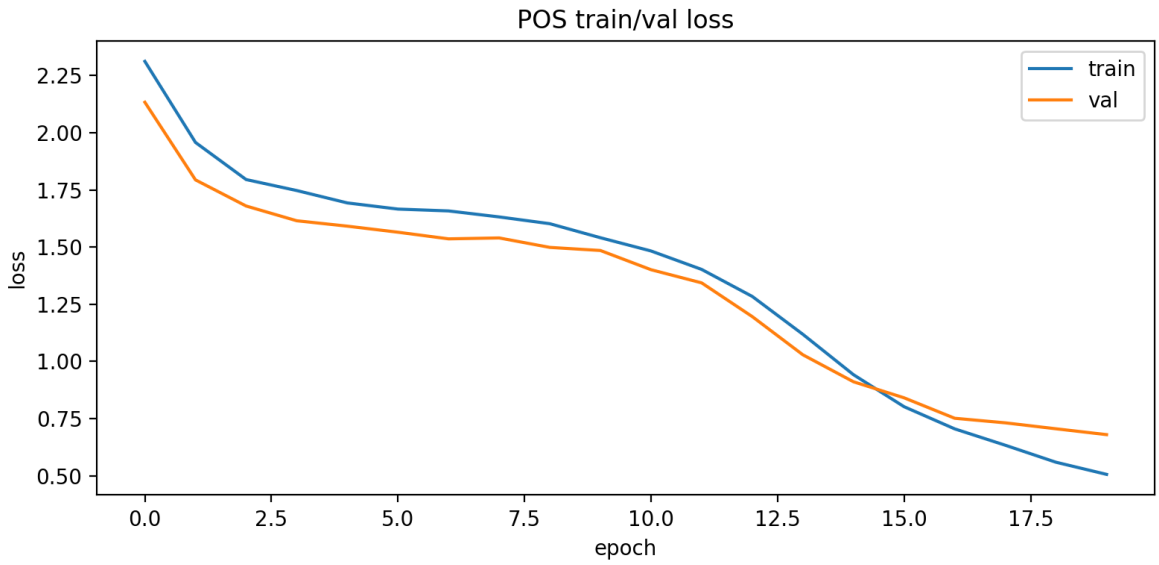
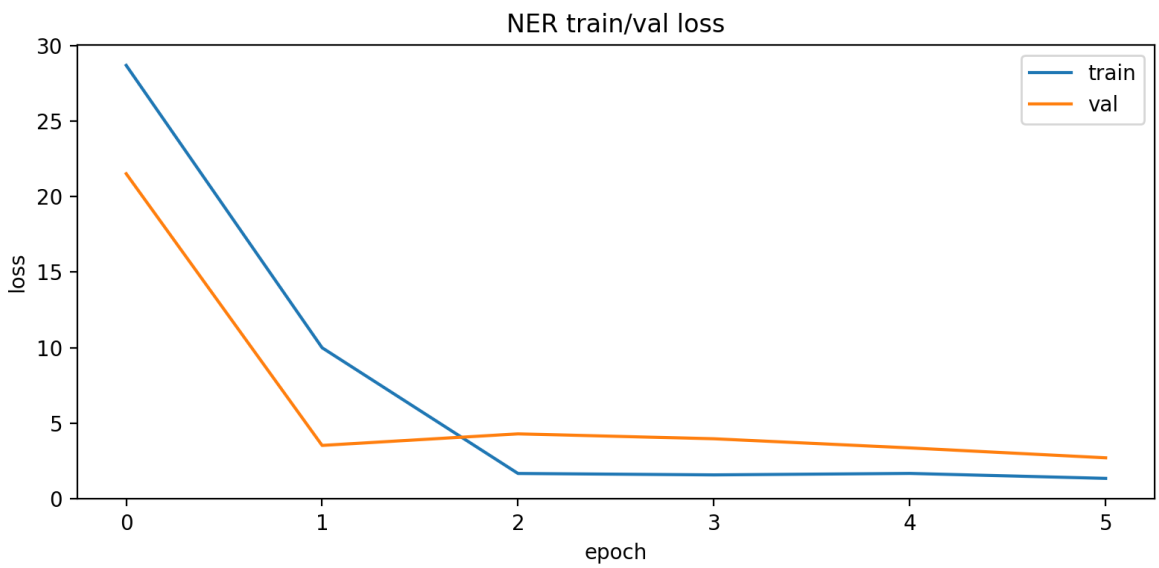


Figure 4. NER train/validation loss



Part 3 Results: Transformer Encoder for Topic Classification

• The classification dataset contained 50 documents, split into train 35, validation 5, and test 10, with balanced class distribution across the five topics.

• A custom Transformer encoder was implemented from scratch using scaled dot-product attention, 4-head self-attention, sinusoidal positional encoding, a position-wise feed-forward network, and 4 stacked Pre-LN encoder blocks. No built-in Transformer classes were used.

• Training remained near chance for the early epochs, then improved sharply after epoch 12. The best validation macro-F1 was 0.7333 at epoch 15.

• On the test set, the Transformer achieved accuracy 0.7000 and macro-F1 0.6267. This is a strong result relative to the very small corpus size and indicates that the document classifier learned a useful topic signal.

• Attention heatmaps were generated successfully for correctly classified examples. Font warnings occurred while displaying Urdu labels in matplotlib, but these are rendering limitations and do not affect the trained model or the correctness of the attention outputs.

Part 3 confusion summary

Confusion matrix: [[2,0,0,0,0],[0,1,1,0,0],[0,0,2,0,0],[2,0,0,0,0],[0,0,0,0,2]]

• Politics, Economy, and Health & Society were classified cleanly in the test set. The main confusion occurred between International and Politics, which is reasonable because both categories share government, diplomacy, and state-related vocabulary.

Figure 5. Transformer training loss

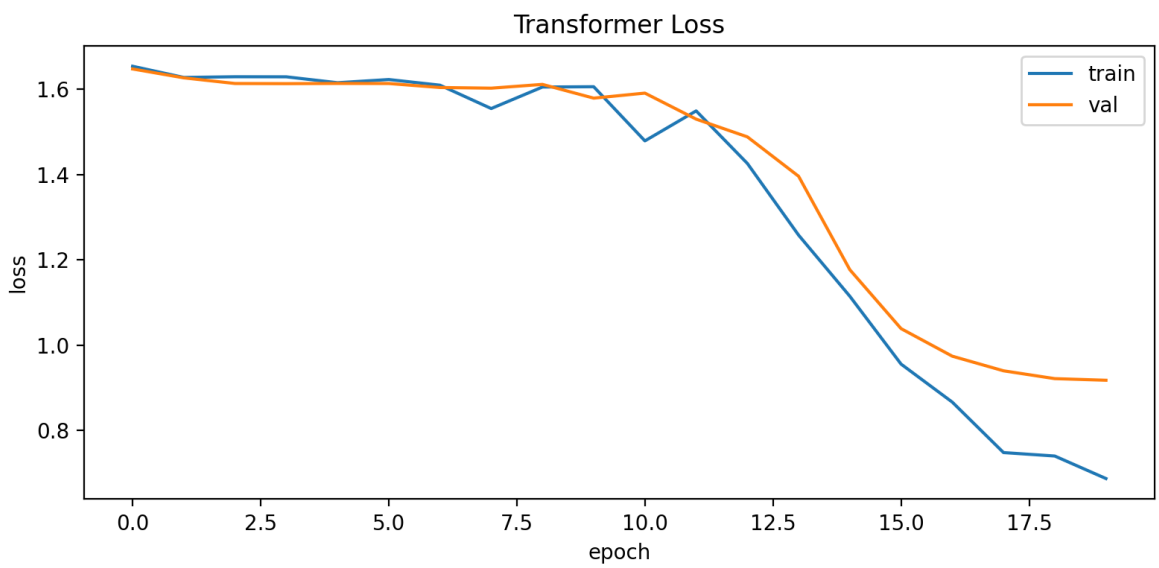
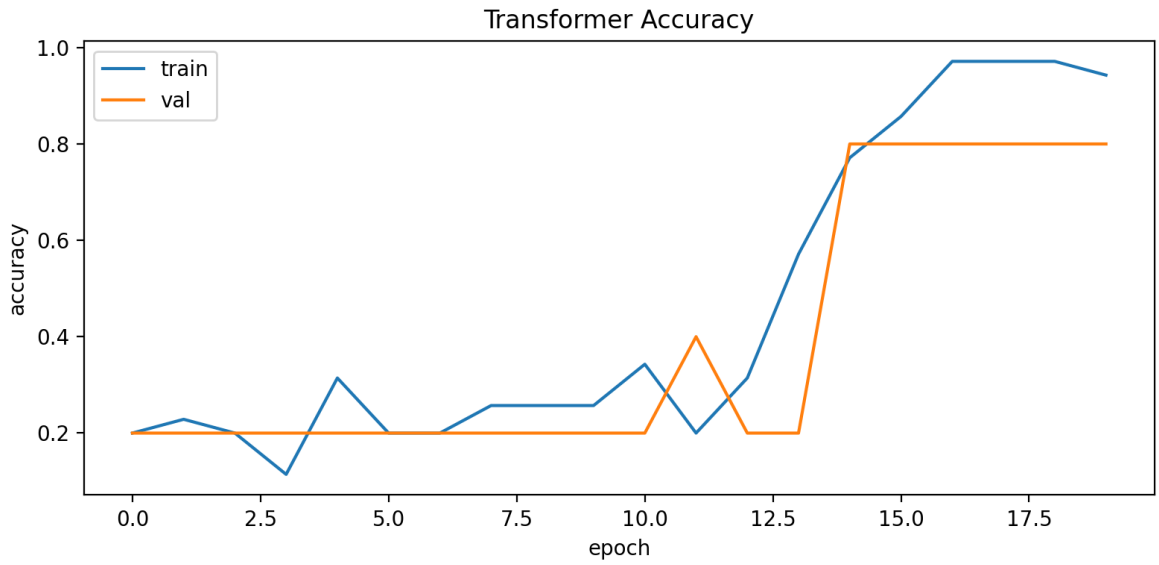


Figure 6. Transformer training accuracy



Conclusion

• The full neural NLP pipeline was implemented successfully from scratch in PyTorch. All required deliverables were produced: weighted lexical matrices, learned embeddings, saved sequence models, saved Transformer classifier, CoNLL files, and training figures.

• The strongest two practical results were POS fine-tuned accuracy 0.7103 / macro-F1 0.4679 and Transformer test accuracy 0.7000 / macro-F1 0.6267.

• The main weakness of the overall project was data scale, not implementation. NER failed mainly because entity labels were too sparse, and Part 1 semantic evaluations were limited by missing words and the small corpus.

• If the corpus were expanded to the intended scale, the same pipeline would likely produce stronger Word2Vec neighbours, more reliable analogies, richer NER supervision, and more stable Transformer generalization.

Appendix: files generated

• embeddings/tfidf_matrix.npy

• embeddings/ppmi_matrix.npy

• embeddings/embeddings_w2v.npy

• models/bilstm_pos.pt

• models/bilstm_ner.pt

• models/transformer_cls.pt

• data/pos_train.conll

- data/pos_test.conll
- data/ner_train.conll
- data/ner_test.conll